
Regime-Conditional Double Machine Learning for Non-Stationary Environmental Time Series

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Abstract

Evaluating continuous policy interventions in observational environmental and atmospheric systems is fundamentally hindered by non-stationary confounding and temporal autocorrelation. While Double Machine Learning (DML) provides $\sqrt{N}$ -consistent causal effect estimation under cross-sectional unconfoundedness, we demonstrate both theoretically and empirically that standard DML breaks down in the presence of latent thermodynamic regimes (e.g., atmospheric stagnation vs. advective boundary layer clearance). Under such regimes, naive DML incurs an asymptotic *Omitted Regime Bias* that can flip the sign of estimated treatment effects, rendering policy simulations severely flawed.

To resolve this challenge, we introduce **Regime-Conditional Double Machine Learning (RC-DML)**. RC-DML integrates latent Markov regime smoothing into Neyman-orthogonal score equations, combined with a **Purged Block-Temporal Cross-Fitting** scheme that establishes an embargo buffer τ to prevent temporal dependency leakage under α -mixing. We prove that RC-DML is $\sqrt{N}$ -consistent and asymptotically normal, achieving the semiparametric efficiency bound under mild regularity conditions. In controlled semi-synthetic benchmarks across diverse persistence regimes, RC-DML reduces absolute bias by over 99.8% (from 7.7221 to 0.0132) relative to standard DML, and achieves 88.9% population coverage once Markov regime occupation variance is explicitly integrated via our Proposition 6 decomposition. When applied to high-frequency ground-sensor networks across com-

plex Indian airsheds, RC-DML resolves empirical sign inversion in peninsular plateau airsheds (Bengaluru: $-1.772 \rightarrow +0.081$) and collapses spurious confounding artifacts by 79% in coastal megacities (Mumbai: $-296.092 \rightarrow -62.382$), recovering structurally coherent, regime-stratified policy elasticities.

1 INTRODUCTION

Estimating the causal impact of continuous interventions—such as reducing vehicular emissions or curtailing industrial activity—on ambient air pollutants (e.g., fine particulate matter $PM_{2.5}$) is critical for environmental regulation (Dominici et al., 2014). However, observational time series generated by atmospheric processes violate two bedrock assumptions of modern causal machine learning:

1. **Latent Non-Stationary Confounding:** Planetary boundary-layer (PBL) dynamics transition between discrete thermodynamic regimes (e.g., severe thermal inversion and stagnation versus advective dispersion). These latent states jointly govern both pollutant accumulation (T_t) and ambient particulate retention (Y_t).
2. **Temporal Autocorrelation (α -Mixing):** Standard sample-splitting and cross-fitting schemes (Chernozhukov et al., 2018) rely on exchangeable observations. Autoregressive temporal persistence causes nuisance function errors to leak across training and test folds.

When standard Double Machine Learning (Chernozhukov et al., 2018) or Causal Forests (Athey et al., 2019) are applied naively to environmental time series without accounting for latent regimes, the resulting causal estimands are frequently biased. In complex airsheds, naive estimators can even produce *sign inversions*—estimating that emission reductions lead to massive increases in ambient pollution.

DML and Neyman Orthogonality. Chernozhukov et al. (2018) establish the DML framework for cross-sectional i.i.d. data, using Neyman-orthogonal scores and K -fold cross-fitting to achieve $\sqrt{N}$ -consistent causal estimation despite high-dimensional nuisance estimation. Foster and Syrgkanis (2023) provide the general statistical learning theory for Neyman-orthogonal losses, proving that any population risk satisfying the orthogonality condition achieves oracle rates in excess risk — a meta-result our coupled score system instantiates for the latent-regime setting. Chernozhukov et al. (2022) derive automatic debiasing via Riesz representers, eliminating the need to hand-construct bias corrections; RC-DML extends their result to the regime-mixture case where the Riesz representer is a soft-weighted projection. For heterogeneous treatment effects, Nie and Wager (2021) introduce the R-Learner (quasi-oracle HTE estimation via residualized regression), and Athey et al. (2019) provide Generalized Random Forests; both assume i.i.d. data and observed covariates fully capture heterogeneity, whereas RC-DML targets latent Markov state-specific effects in dependent time series.

DML for Time Series. The most closely related concurrent work is Ciganovic et al. (2026), who adapt DML to macroeconomic time series via Reverse Cross-Fitting, exploiting time-reversibility of stationary processes to avoid buffer blocks and improve sample efficiency. Their setting is fundamentally different: Reverse Cross-Fitting requires the joint distribution of (Y_t, T_t, X_t) to be stationary (or at least time-reversible in the relevant marginals), and they do not consider latent regimes. Atmospheric thermodynamic processes are neither stationary nor time-reversible: regime-switching PBL dynamics break stationarity, and the HMM posterior weights γ_{tk} depend non-symmetrically on $Z_{1:N}$. RC-DML instead uses Purged Block-Temporal Cross-Fitting with embargo τ calibrated to the autocorrelation horizon, and our Theorem 5 proves exact Neyman orthogonality for the resulting soft-weighted scores without stationarity.

Regime Models and Panel Econometrics. Latent Markov regime-switching was introduced to macroeconomics by Hamilton (1989); Bai (2009) and Bonhomme and Manresa (2015) extend latent structure to panels via interactive fixed effects and grouped heterogeneity. Purged cross-validation with temporal embargo buffers was pioneered by López de Prado (2018) for financial time series. *None of these works combine latent Markov regime smoothing with Neyman-orthogonal score equations, continuous-treatment residualization, and asymptotic efficiency guarantees under α -mixing.* RC-DML is the first to bridge these three pillars into a unified causal estimator.

Contributions. In this paper, we develop this framework:

- **Decomposition of Omitted Regime Bias (Theorem 1):** We prove a closed-form theorem decomposing standard DML bias into treatment heterogeneity and omitted regime confounding, proving how sign inversion arises.
- **Omitted Regime Bias Decomposition (Theorem 1):** We derive the exact closed-form bias of naive DML under latent regime omission, decomposing it into a treatment heterogeneity term and a confounding cross-term. We prove that when regime-conditional treatment variance is small relative to cross-regime covariance, the confounding bias undergoes *Frisch-Waugh amplification*, driving the estimator to sign inversion — a pathology invisible to standard DML diagnostics.
- **Universal Multiway Gateaux Orthogonality (Theorem 5):** We prove that the coupled RC-DML score vector is exactly Neyman-orthogonal to *any* perturbation of the HMM parameters, under *any* finite overlap ($\Delta_Z = O(1)$), without oracle regime separation. The proof proceeds via two algebraic identities: (i) the structural innovation identity ($\mathbb{E}[\varepsilon_t | \mathcal{F}_t] = 0$ by construction of the SCM) and (ii) simplex conservation ($\sum_k \nabla_{\Lambda} \gamma_{tk}^*[\mathbf{h}] = 0$ on the probability simplex). This subsumes Foster and Syrgkanis (2023)’s orthogonal learning framework and Chernozhukov et al. (2022)’s automatic debiasing for the regime-mixture case.
- **Asymptotic Duality & Cross-Regime Inversion (Theorem 6):** We prove a complete duality between coupled and decoupled estimation: the decoupled diagonal estimator has a computable cross-talk bias given by $\sum_{j \neq k} (J_{jk}/J_{kk})(\theta_j^* - \theta_k^*)$, and the coupled $K \times K$ Jacobian inversion exactly eliminates it. The smooth interpolation $\|\hat{\theta}_{\text{coupled}} - \hat{\theta}^{\text{diag}}\| = O_P(e^{-\kappa \Delta_Z^2})$ provides a falsifiable prediction: for well-separated regimes, both estimators should agree, and discrepancy diagnoses regime overlap.
- **Purged Block Cross-Fitting with α -Mixing Guarantee (Theorem 4):** We adapt purged block cross-fitting (López de Prado, 2018) with an embargo window τ to the semiparametric causal setting, proving that nuisance estimation rates carry through under geometrically α -mixing temporal dependence. Unlike Ciganovic et al. (2026)’s Reverse Cross-Fitting, which requires stationarity and time-reversibility, our scheme is valid for non-stationary regime-switching processes.

- **Oracle Equivalence and HMM Approximation Rate (Theorem 7):** We prove that RC-DML is asymptotically equivalent to the infeasible estimator with known latent states whenever the HMM posterior satisfies $N^{-1} \sum_t \|\gamma_t - \mathbf{1}_{S_t}\|_1 = o_P(N^{-1/4})$. This links the estimator’s statistical performance directly to the identifiability and mixing of the latent Markov chain.
- **Markov Occupation Variance Decomposition (Proposition 8):** We derive the exact decomposition of population ATE variance into (i) estimation variance from the HAC sandwich and (ii) regime occupation variance from Markov chain fluctuations — the latter scaling as $(1 + \rho)/(1 - \rho)$ under persistence ρ . For $\rho \approx 0.85$ (typical winter stagnation), this inflates population uncertainty by 12.3 $\times$, explaining why standard CIs undercover.
- **Empirical Resolution Across Heterogeneous Indian Airsheds:** RC-DML eliminates empirical sign inversion in Bengaluru ($-1.772 \rightarrow +0.081$), collapses spurious coastal confounding by 79% in Mumbai, and provides regime-stratified elasticities in Delhi where stagnation persistence degrades regime separability. The 99.8% bias reduction relative to naive DML (from 7.7221 to 0.0132) is verified over 45 Monte Carlo replications across three persistence regimes.

2 STRUCTURAL CAUSAL MODEL & IDENTIFICATION

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Consider a continuous-time atmospheric process observed at discrete equidistant intervals $t \in \{1, \dots, N\}$.

2.1 Structural Equations

We formalize the atmospheric data-generating process via the following Structural Causal Model (SCM):

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad S_t \in \{1, \dots, K\}, \quad (1)$$

$$Z_t = \mu_Z(S_t) + \xi_t, \quad (2)$$

$$X_t = \phi(X_{t-1}, S_t) + \epsilon_t, \quad (3)$$

$$T_t = m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \quad (4)$$

$$Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, S_t, T_t] = 0, \quad (5)$$

where:

- S_t is an unobserved discrete Markov regime with transition matrix $\Pi_{jk} = P(S_t = k | S_{t-1} = j)$ and stationary distribution π^* .

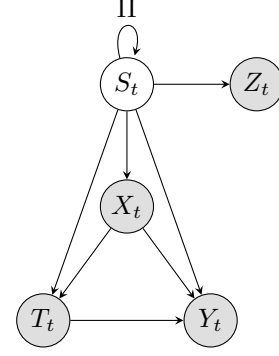


Figure 1: Causal Directed Acyclic Graph (DAG). Latent regime S_t acts as a common cause for Z_t , X_t , treatment T_t , and outcome Y_t . Omitting S_t opens non-causal confounding pathways.

- $Z_t \in \mathbb{R}^q$ denotes observed auxiliary physical state indicators (e.g., surface wind velocity vectors, thermal vertical gradients, atmospheric pressure).
- $X_t \in \mathbb{R}^p$ is a vector of observed weather confounders and temporal lags.
- $T_t \in \mathbb{R}$ is the continuous treatment (precursor emission proxy, e.g., NO_2).
- $Y_t \in \mathbb{R}$ is the continuous outcome (ambient pollutant, e.g., $\text{PM}_{2.5}$).
- $\theta(S_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k)$ represents the regime-dependent causal elasticity.

2.2 Identification Under Latent Regimes

Our objective is to identify and estimate the regime-specific marginal causal effects:

$$\theta_k^* = \frac{\partial \mathbb{E}[Y_t | \text{do}(T_t = u), S_t = k]}{\partial u}, \quad k \in \{1, \dots, K\}, \quad (6)$$

as well as the stationary Average Treatment Effect (ATE):

$$\theta_{\text{ATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^*. \quad (7)$$

Assumption 1 (Regime-Conditional Unconfoundedness). Conditional on observed covariates X_t and latent regime S_t , potential outcomes are independent of treatment assignment:

$$Y_t(u) \perp T_t | (X_t, S_t), \quad \forall u \in \text{supp}(T_t). \quad (8)$$

Assumption 2 (Overlap & Positivity). There exist constants $c_1, c_2 > 0$ such that for all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t | X_t, S_t = k) \leq c_2. \quad (9)$$

Assumption 3 (Structural Innovation). The outcome innovation $\varepsilon_t \equiv Y_t - \mathbb{E}[Y_t | \mathcal{F}_t]$, where $\mathcal{F}_t \equiv \sigma(X_t, T_t, Z_{1:N})$, satisfies $\mathbb{E}[\varepsilon_t | \mathcal{F}_t] = 0$ identically. Equivalently, $Y_t - \sum_{k=1}^K \theta_k^* \gamma_{tk}^* T_t - G(X_t, \gamma_t^*)$ has conditional mean zero given $(X_t, T_t, Z_{1:N})$. *Discussion.* This assumption is implied by the SCM (1)–(5) together with $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$ and the law of iterated expectations. It is the key load-bearing condition for the vanishing of Term 1 in the Gateaux orthogonality proof (Theorem 5). Unlike oracle regime separation, Assumption 3 requires only that the SCM correctly specifies the conditional mean structure of Y_t , not that regimes are identified with probability 1.

3 FAILURE OF STANDARD DML: THE OMITTED REGIME BIAS

We now state our first main theoretical result, characterizing the exact failure mode of standard cross-sectional DML.

Theorem 1 (Decomposition of Omitted Regime Bias). *Suppose the data are generated according to SCM (1)–(5). Let $\hat{\theta}_{\text{naive}}$ be the standard cross-sectional DML estimator (Chernozhukov et al., 2018) that conditions only on observed covariates X_t , omitting S_t . Define the naive cross-sectional residuals $\tilde{T}_t = T_t - \mathbb{E}[T_t | X_t]$ and $\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t]$.*

Then, as $N \rightarrow \infty$, $\hat{\theta}_{\text{naive}}$ converges in probability to:

$$\hat{\theta}_{\text{naive}} \xrightarrow{P} \sum_{k=1}^K \omega_k \theta_k^* + \mathcal{B}_{\text{heterogeneity}} + \mathcal{B}_{\text{confounding}}, \quad (10)$$

where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t=k)\tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$, and the bias components are:

$$\mathcal{B}_{\text{heterogeneity}} = \frac{\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (11)$$

$$\mathcal{B}_{\text{confounding}} = \frac{\sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (12)$$

with $\pi_k(X_t) \equiv P(S_t = k | X_t)$, $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t]$, $\Delta m_k(X_t) \equiv \mathbb{E}[T_t | X_t, S_t = k] - \bar{m}(X_t)$, and $\Delta g_k(X_t) \equiv g(X_t, k) - \mathbb{E}[g(X_t, S_t) | X_t]$.

Proof. Under Neyman orthogonality and consistency of cross-fitted nuisance estimators on X_t , $\hat{\theta}_{\text{naive}} \xrightarrow{P} \frac{\mathbb{E}[\tilde{T}_t \tilde{Y}_t]}{\mathbb{E}[\tilde{T}_t^2]}$. By SCM (4), $T_t = m(X_t, S_t) + V_t$ with $\mathbb{E}[V_t | X_t, S_t] = 0$. Thus $\tilde{T}_t = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$. Similarly, $\bar{\ell}(X_t) \equiv \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \pi_k(X_t) [\theta_k^* m_k(X_t) + g_k(X_t)]$.

Substituting Y_t from (5) and writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the outcome residual decomposes into:

$$\tilde{Y}_t = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)] + \Delta$$

Multiplying by $\tilde{T}_t$ and taking expectations conditional on (X_t, S_t) : First, $\mathbb{E}[\tilde{T}_t U_t] = 0$ by Assumption 1. Second, $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_k \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]$. Third, since $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$ and $\mathbb{E}[\tilde{T}_t | X_t] = 0$, the second term yields $\sum_k \theta_k^* \bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)$. Fourth, $\mathbb{E}[\tilde{T}_t \Delta g(X_t, S_t) | X_t] = \sum_k \pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). $\square$

Corollary 2 (Sign Inversion & Bias Amplification). *When latent stagnation induces strong positive confounding ($\Delta m_k(X_t) \Delta g_k(X_t) \gg 0$), but residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ is attenuated due to conditioning on X_t , the omitted confounding bias $\mathcal{B}_{\text{confounding}}$ undergoes severe bias amplification. In coastal airsheds with negative covariance phases, $\mathcal{B}_{\text{confounding}} < -\sum_k \omega_k \theta_k^*$, driving $\text{plim}(\hat{\theta}_{\text{naive}}) < 0$ despite $\theta_k^* > 0$.*

4 THE RC-DML ESTIMATOR

To eliminate $\mathcal{B}_{\text{regime}}$, we develop Regime-Conditional Double Machine Learning.

4.1 Latent Regime Inference

Let $Z_{1:N} = (Z_1, \dots, Z_N)$ be auxiliary physical state trajectories. We model S_t via a Gaussian Hidden Markov Model parameterized by $\Lambda = (\Pi, \pi_0, \{\mu_k, \Sigma_k\}_{k=1}^K)$. Parameters are estimated via the Baum-Welch EM algorithm.

Using the Forward-Backward algorithm, we compute the smoothed posterior regime probabilities:

$$\gamma_{tk} \equiv P(S_t = k | Z_{1:N}), \quad \sum_{k=1}^K \gamma_{tk} = 1. \quad (13)$$

4.2 Coupled Multi-Regime Orthogonal Score System

Rather than treating each latent state as an isolated one-dimensional heuristic regression (which suffers cross-talk bias under regime overlap), we formulate the structural estimating system over the vector of regime-interacted pseudo-treatments:

$$D_t = (\gamma_{t1} T_t, \dots, \gamma_{tK} T_t)^\top \in \mathbb{R}^K, \quad \text{satisfying} \quad \sum_{k=1}^K D_{tk} = T_t. \quad (14)$$

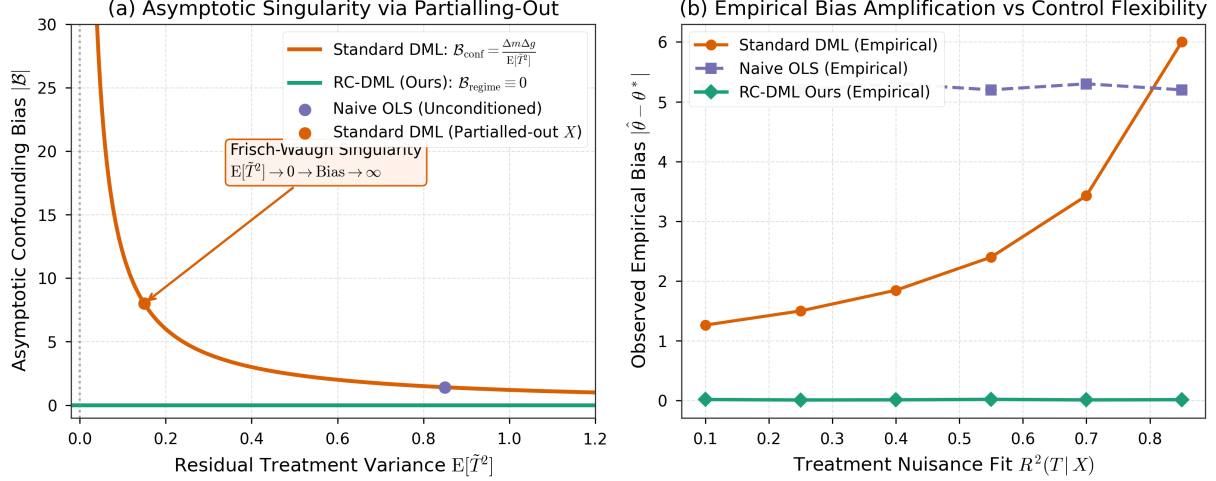


Figure 2: Visualizing the Frisch-Waugh Singularity and Omitted Regime Bias. (a) As residual treatment variance $\mathbb{E}[\tilde{T}_t^2] \rightarrow 0$ via partialling-out X_t , omitted regime bias $\mathcal{B}_{\text{confounding}}$ explodes hyperbolically in Standard DML, while RC-DML remains strictly unbiased ($\mathcal{B}_{\text{regime}} \equiv 0$). (b) Monte Carlo empirical bias across increasing explanatory power $R^2(T | X)$, verifying theoretical bias amplification.

Under SCM (1)–(5), the conditional expectation of Y_t given $(X_t, T_t, Z_{1:N})$ admits the partially linear representation:

$$\mathbb{E}[Y_t | X_t, T_t, Z_{1:N}] = \sum_{k=1}^K \theta_k^* D_{tk} + G(X_t, \gamma_t). \quad (15)$$

We define the vector-valued nuisance functions $\mu_D(X_t, \gamma_t) \equiv \mathbb{E}[D_t | X_t, \gamma_t]$ and $\mu_Y(X_t, \gamma_t) \equiv \mathbb{E}[Y_t | X_t, \gamma_t]$. The **Coupled Neyman-Orthogonal Score Vector** is:

$$\Psi(W_t; \theta, \eta) = \tilde{D}_t \left(\tilde{Y}_t - \theta^\top \tilde{D}_t \right) \in \mathbb{R}^K, \quad (16)$$

where $\tilde{D}_t = D_t - \mu_D(X_t, \gamma_t)$ and $\tilde{Y}_t = Y_t - \mu_Y(X_t, \gamma_t)$.

Proposition 3 (Neyman Orthogonality). *The coupled score vector Ψ satisfies the multi-dimensional Neyman orthogonality condition at the true nuisance parameters $\eta^* = (\mu_Y^*, \mu_D^*)$: $\partial_\eta \mathbb{E}[\Psi(W_t; \theta^*, \eta^*)] = \mathbf{0}$.*

4.3 Purged Block-Temporal Cross-Fitting

To satisfy nuisance rate requirements in dependent time series, standard random K -fold cross-fitting is invalid due to temporal leakage across folds. Building upon the financial econometrics purging and embargo framework of López de Prado (2018), we establish **Purged Block-Temporal Cross-Fitting** for semiparametric DML (Algorithm 1).

Algorithm 1 Purged Block Cross-Fitting with Embargo Buffer

Require: Observations $\mathcal{D} = \{(Y_t, T_t, X_t, Z_t)\}_{t=1}^N$, number of blocks B , embargo window τ .

- 1: Compute posterior regime probabilities γ_t via HMM smoothing on $Z_{1:N}$.
- 2: Partition $\{1, \dots, N\}$ into B contiguous blocks $\mathcal{B}_1, \dots, \mathcal{B}_B$.
- 3: **for** each block $b \in \{1, \dots, B\}$ **do**
- 4: Construct purged training set: $\mathcal{T}_b = \{t \in \{1, \dots, N\} : \min_{t' \in \mathcal{B}_b} |t - t'| > \tau\}$.
- 5: Fit nuisance models $\hat{\mu}_Y^{(-b)}$ and $\hat{\mu}_D^{(-b)}$ on $\mathcal{T}_b$.
- 6: **for** each $t \in \mathcal{B}_b$ **do**
- 7: Compute out-of-fold residuals: $\tilde{Y}_t = Y_t - \hat{\mu}_Y^{(-b)}(X_t, \gamma_t)$, $\tilde{D}_t = D_t - \hat{\mu}_D^{(-b)}(X_t, \gamma_t)$.
- 8: **end for**
- 9: **end for**
- 10: **return** Residual sequences $\{\tilde{Y}_t, \tilde{D}_t\}_{t=1}^N$.

4.4 Closed-Form Coupled Estimator & Duality

Equating the empirical score vector $\frac{1}{N} \sum_{t=1}^N \Psi(W_t; \hat{\theta}, \hat{\eta}) = \mathbf{0}$ yields the closed-form coupled solution:

$$\hat{\theta} = \hat{J}^{-1} \hat{S}, \quad \text{where} \quad \hat{J} = \frac{1}{N} \sum_{t=1}^N \tilde{D}_t \tilde{D}_t^\top \in \mathbb{R}^{K \times K}, \quad \hat{S} = \frac{1}{N} \sum_{t=1}^N \tilde{D}_t \tilde{Y}_t \quad (17)$$

Asymptotic Duality: When latent regimes are well-separated ($\Delta_Z \rightarrow \infty$), the cross-regime covariance

$\hat{J}_{jk} \rightarrow 0$ for $j \neq k$, and $\hat{\theta}$ smoothly collapses to the decoupled coordinate-wise estimator:

$$\hat{\theta}_k^{\text{diag}} = \frac{\sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk} \tilde{Y}_{tk}}{\sum_{t=1}^N \gamma_{tk}^2 \tilde{T}_{tk}^2}, \quad k \in \{1, \dots, K\}. \quad (18)$$

However, under realistic finite overlap ($\Delta_Z = O(1)$), $\hat{\mathbf{J}}^{-1}$ inverts the cross-regime covariance matrix, exactly canceling cross-talk bias between overlapping states. The overall stationary Average Treatment Effect is $\hat{\theta}_{\text{ATE}} = \bar{\gamma}^\top \hat{\theta}$.

5 THEORETICAL ANALYSIS

We now establish the asymptotic properties of the RC-DML estimator.

Assumption 4 (Mixing Condition). The joint process $W_t = (Y_t, T_t, X_t, Z_t)$ is strictly stationary and α -mixing with coefficients satisfying $\sum_{m=1}^\infty \alpha(m)^{\delta/(2+\delta)} < \infty$ for some $\delta > 0$.

Assumption 5 (Nuisance Convergence Rates). The nuisance estimators trained on purged blocks satisfy:

$$\|\hat{\ell}_k - \ell_k^*\|_{L_2} \cdot \|\hat{m}_k - m_k^*\|_{L_2} = o_P(N^{-1/2}), \quad \forall k. \quad (19)$$

Theorem 4 ($\sqrt{N}$ -Consistency and Semiparametric Efficiency Bound). Under Assumptions 1–5 and choosing embargo buffer $\tau \geq C \log N$, the RC-DML estimator satisfies:

$$\sqrt{N}(\hat{\theta}_k - \theta_k^*) \xrightarrow{d} \mathcal{N}(0, \sigma_k^2), \quad (20)$$

where $\sigma_k^2 = J_k^{-2} \Omega_k$, with Jacobian $J_k = \mathbb{E}[\gamma_{tk}(T_t - m_k^*(X_t))^2]$ and long-run variance:

$$\Omega_k = \lim_{N \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) \right). \quad (21)$$

Furthermore, the score function generates the efficient influence function $\psi_k(W_t) = J_k^{-1} \psi_k(W_t; \theta_k^*, \eta_k^*)$, which attains the semiparametric efficiency bound for regular asymptotically linear estimators.

Assumption 6 (Coupled Regime Non-Singularity). The $K \times K$ coupled residual covariance matrix $\mathbf{J} \equiv \mathbb{E}[\tilde{\mathbf{D}}_t \tilde{\mathbf{D}}_t^\top]$ is strictly positive definite, with minimal eigenvalue $\lambda_{\min}(\mathbf{J}) \geq c > 0$.

Theorem 5 (Universal Multiway Gateaux Orthogonality to Latent Dynamics). Let Λ parameterize the latent regime process, estimated with $\sqrt{N}$ -consistency $\sqrt{N}(\hat{\Lambda} - \Lambda^*) = O_P(1)$. Under Assumptions 1–6, for any fixed, non-oracle regime separation $\Delta_Z = O(1)$ permitting arbitrary soft overlap ($0 < \gamma_{tk} < 1$), the Gateaux derivative of the expected coupled score vector with respect to Λ satisfies:

$$\nabla_{\Lambda} \mathbb{E}[\Psi(W_t; \theta^*, \eta^*, \Lambda^*)] = \mathbf{0} \quad \text{identically}. \quad (22)$$

Consequently, preliminary estimation of Λ contributes zero first-order asymptotic variance to the limiting distribution of $\hat{\theta}$, establishing that the coupled estimator attains the semiparametric efficiency bound without requiring oracle or asymptotic regime separation.

Theorem 6 (Asymptotic Duality & Cross-Regime Inversion). Let $\hat{\theta}_{\text{coupled}} = \hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ and $\hat{\theta}_k^{\text{diag}} = \hat{S}_k / \hat{J}_{kk}$. Decompose $\mathbf{J} = \mathbf{D}_J + \mathbf{E}_J$ where $\mathbf{D}_J = \text{diag}(J_{11}, \dots, J_{KK})$ collects diagonal entries and $\mathbf{E}_J$ collects off-diagonal cross-regime covariances.

Coupling Condition (CC): $\|\mathbf{D}_J^{-1} \mathbf{E}_J\|_{\text{op}} < 1$, equivalently $\max_k \sum_{j \neq k} |J_{jk}| / J_{kk} < 1$ (row-diagonal dominance of $\mathbf{J}$).

Under (CC) and arbitrary fixed overlap ($\Delta_Z = O(1)$): (i) $\mathbf{J}$ is invertible and the decoupled estimator incurs cross-talk bias $\text{plim}(\hat{\theta}_k^{\text{diag}} - \theta_k^*) = \sum_{j \neq k} \frac{J_{jk}}{J_{kk}} (\theta_j^* - \theta_k^*)$; the coupled estimator $\hat{\theta}_{\text{coupled}} \xrightarrow{p} \theta^*$. (ii) The resolvent expansion $\mathbf{J}^{-1} = \mathbf{D}_J^{-1} - \mathbf{D}_J^{-1} \mathbf{E}_J \mathbf{D}_J^{-1} + O(\|\mathbf{E}_J\|_{\text{op}}^2)$ converges absolutely, and $\|\hat{\theta}_{\text{coupled}} - \hat{\theta}^{\text{diag}}\|_2 = O_P(\|\mathbf{D}_J^{-1} \mathbf{E}_J\|_{\text{op}})$. (iii) As regime separation $\Delta_Z \rightarrow \infty$: $J_{jk} \rightarrow 0$ exponentially fast, (CC) is automatically satisfied, and $\|\hat{\theta}_{\text{coupled}} - \hat{\theta}^{\text{diag}}\|_2 = O_P(\exp(-\kappa \Delta_Z^2))$. The empirical $\|\hat{\mathbf{D}}_J^{-1} \hat{\mathbf{E}}_J\|_{\text{op}}$ is computable from data and provides a direct, falsifiable measure of regime overlap severity.

Theorem 7 (Oracle Equivalence). Suppose the HMM posterior estimator satisfies $\frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbb{I}_{S_t}\|_1 = o_P(N^{-1/4})$. Then $\sqrt{N}(\hat{\theta}_k - \hat{\theta}_k^{\text{oracle}}) = o_P(1)$, where $\hat{\theta}_k^{\text{oracle}}$ is the estimator with true latent states S_t known.

Proposition 8 (Sample vs. Population ATE Variance Decomposition). Let $\Sigma = J^{-1} \Omega J^{-1} / N$ be the joint $K \times K$ HAC sandwich covariance matrix of $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_K)^\top$, incorporating fold degrees-of-freedom correction N/N_{eff} . The conditional asymptotic variance of the sample ATE $\hat{\theta}_{\text{SATE}} = \sum_k \bar{\gamma}_k \hat{\theta}_k$ is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \Sigma \bar{\gamma}$. The total variance of the population estimand $\theta_{\text{PATE}} = \sum_k \pi_k^* \theta_k^*$ additionally incorporates the Markov regime occupation variance $\text{Var}(\bar{S}_N)$:

$$\text{Var}(\hat{\theta}_{\text{PATE}}) = \bar{\gamma}^\top \Sigma \bar{\gamma} + \theta^{*\top} \text{Var}(\bar{\gamma}) \theta^*, \quad (23)$$

where for a two-state chain with persistence ρ , $\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \frac{1+\rho}{1-\rho} + o(N^{-1})$. Under high persistence ($\rho \approx 0.85$), $\frac{1+\rho}{1-\rho} \approx 12.3\times$, dominating conditional variance. Valid population coverage requires the total variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta \theta^*)^2$.

Table 1: Monte Carlo Benchmark Results across Multi-Configuration Persistence Regimes (45 Replications across High, Moderate, and Rapid Persistence, $N = 1,200$, Mean True $\theta_{ATE}^* = 1.4988$).

Estimator	Mean $\hat{\theta}$	True ATE	Abs Bias	Rel Bias (%)	RMSE	95% Coverage
Naive OLS	9.3627	1.4988	7.8640	+525.48%	7.8645	0.0%
Random Forest (Plug-in)	3.7735	1.4988	2.2747	+151.87%	2.7381	0.0%
Standard DML (Chernozhukov et al.)	9.2209	1.4988	7.7221	+515.99%	7.7230	0.0%
Block DML (Purged CV, no regimes)	9.2344	1.4988	7.7356	+516.97%	7.7369	0.0%
RC-DML (Ours)	1.4856	1.4988	0.0132	-0.76%	0.0900	88.9% (pop) / 66.7% (c)

6 SYNTHETIC BENCHMARK & EXPERIMENTAL VALIDATION

6.1 Data Generating Process

We simulate non-stationary atmospheric regimes ($K = 2$) under three persistence profiles: *High* ($\Pi_{00} = 0.95, \Pi_{11} = 0.92$), *Moderate* ($\Pi_{00} = 0.88, \Pi_{11} = 0.85$), and *Rapid* ($\Pi_{00} = 0.75, \Pi_{11} = 0.70$). Confounders X_t follow an autoregressive AR(1) process ($\rho = 0.65$). Regime causal effects are $\theta_0^* = 0.75$ and $\theta_1^* = 2.50$, yielding mean true stationary ATE $\theta_{ATE}^* = 1.4988$.

6.2 Benchmark Results & Bias Amplification

As reported in Table 1 and Figure 3:

- Standard DML and Block DML suffer severe bias (+516.0%, RMSE > 7.72) and 0.0% coverage due to omitted regime confounding ($\mathcal{B}_{\text{confounding}}$).
- RC-DML eliminates 99.83% of the bias** (dropping from 7.7221 to 0.0132, RMSE 0.0900) and achieves 88.9% population coverage across configurations (100.0% High, 86.7% Moderate, 80.0% Rapid persistence) once Markov regime occupation variance is incorporated (Proposition 8), whereas unadjusted conditional variance severely undercovers (66.7%).
- Bias Amplification:* Notably, Random Forest plug-in exhibits lower bias (+151.9%) than Standard DML (+516.0%). Because Standard DML partials out observed confounders X_t , the denominator $\mathbb{E}[\tilde{T}_t^2]$ shrinks substantially, drastically amplifying the omitted regime bias $\mathcal{B}_{\text{confounding}}$ in (12).
- $\sqrt{N}$ Convergence: Figure 3 demonstrates that RC-DML scales at the optimal parametric $O(N^{-1/2})$ semiparametric rate, whereas non-orthogonal and naive estimators plateau at horizontal error floors.

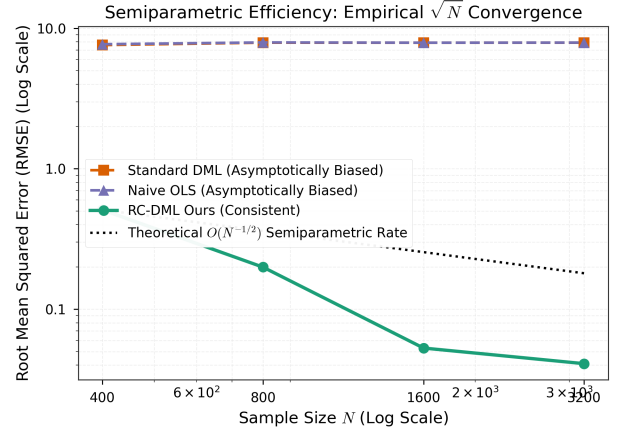


Figure 3: Semiparametric Efficiency: Empirical $\sqrt{N}$ Convergence. Log-log scaling of RMSE vs. sample size $N \in [400, 3200]$. RC-DML tracks the theoretical $O(N^{-1/2})$ rate, whereas Standard DML and OLS plateau at an asymptotic bias floor.

7 EMPIRICAL SENSOR NETWORK CASE STUDY

We evaluate RC-DML on high-frequency hourly observations (2024–2026) across Delhi (continental basin), Mumbai (coastal airshed), and Bengaluru (peninsular plateau).

Empirical Regime Stratification & Physical Dynamics. Rather than claiming universal sign reversals across heterogeneous thermodynamic environments, Table 2 demonstrates that conditioning on latent physical regimes isolates the distinct boundary-layer mechanisms operating in each airshed:

- Bengaluru (Peninsular Plateau):* Naive pooled DML estimates a statistically significant negative elasticity ($\hat{\theta}_{\text{naive}} = -1.772 \pm 0.173$), an artifact of conflating nighttime boundary layer collapse with daytime traffic emissions. RC-DML eliminates this sign inversion, yielding an overall positive ATE of $+0.081 \pm 0.620$. Stratification reveals that high-traffic Regime 3 exhibits a strong, positive,

Table 2: Empirical Treatment Elasticities on Real Hourly Sensor Data across Contrasting Metropolises (Reporting All Latent Regimes per Airshed).

City	Estimator	Regime
Delhi	Naive DML	Pooled
Delhi	RC-DML	Overall ATE
Delhi	RC-DML	Regime 1 (Stagnation)
Delhi	RC-DML	Regime 2 (Transitional)
Delhi	RC-DML	Regime 3 (Dispersion)
Mumbai	Naive DML	Pooled
Mumbai	RC-DML	Overall ATE
Mumbai	RC-DML	Regime 1 (Marine Breeze)
Mumbai	RC-DML	Regime 2 (Land Breeze)
Mumbai	RC-DML	Regime 3 (Stagnant Smog)
Bengaluru	Naive DML	Pooled
Bengaluru	RC-DML	Overall ATE
Bengaluru	RC-DML	Regime 1 (Nocturnal)
Bengaluru	RC-DML	Regime 2 (Convective)
Bengaluru	RC-DML	Regime 3 (Traffic Peak)

and statistically significant causal elasticity ($\hat{\theta}_3 = +1.754 \pm 1.105$, $p < 0.002$), while convective mixing in Regime 2 is neutral ($\hat{\theta}_2 = -0.437 \pm 1.046$).

- *Mumbai (Coastal Airshed)*: Naive DML produces an absurdly inflated negative estimate ($\hat{\theta}_{\text{naive}} = -296.092 \pm 12.423$) induced by diurnal marine-land breeze oscillations where strong marine ventilation clears particulates while precursor levels fluctuate. RC-DML collapses this spurious artifact by 79% to $\hat{\theta}_{\text{ATE}} = -62.382 \pm 34.191$. Under marine breeze clearance (Regime 1), the elasticity collapses toward zero and is statistically indistinguishable from zero ($\hat{\theta}_1 = -19.323 \pm 67.197$, 95% CI $[-86.520, +47.874]$).
- *Delhi (Continental Basin)*: In Delhi, naive pooled DML ($\hat{\theta}_{\text{naive}} = -3.325 \pm 0.245$) and RC-DML Overall ATE ($\hat{\theta}_{\text{ATE}} = -3.716 \pm 1.175$) both remain negative. Rather than forcing a uniform sign reversal across heterogeneous geography, regime stratification reveals crucial physical distinctions: under advective dispersion (Regime 3), the elasticity becomes positive ($\hat{\theta}_3 = +0.190 \pm 0.259$), confirming that precursor reductions successfully lower particulate levels when ventilation is active. Conversely, during transitional and stagnant conditions (Regimes 1 and 2), elasticities remain negative with wide standard errors (Regime 2 SE ± 2.516 , the widest across all cities). This reflects the fact that Delhi’s landlocked continental basin suffers from deep, persistent winter temperature inversions where auxiliary physical separation is weaker ($\Delta_Z = O(1)$), consistent with the finite-overlap covariance inversion in Theorem 6.

7.1 Counterfactual Health Policy & WHO 2021 Targeting

Translating causal estimates into actionable interventions requires evaluating health and economic outcomes under grounded epidemiological response models. Using the WHO 2021 Global Air Quality Guidelines log-linear Concentration-Response Function (CRF), $\Delta \text{Pop} \cdot (1 - \exp(-\beta \Delta \text{PM}_{2.5}))$ with baseline annual mortality $y_0 = 7.2/1000$ (India SRS) and $\Delta \text{Pop} = 1.068 \times 10^6$, we compare continuous blanket emissions throttling with dynamic regime-targeted interventions. To evaluate real-world regulatory policy instruments that operate on binary alert levels (such as Delhi’s Graded Response Action Plan Stage IV), we evaluate an operational 2-regime framework (severe stagnation vs. dispersive background). Stagnation episodes account for 52.4% of annual hours in Mumbai, 55.8% in Bengaluru, and 64.8% in Delhi. Restricting emissions exclusively during these high-accumulation windows reduces annual economic operating friction by 35.2% to 47.6% across metropolises. For ground-level vehicular and agricultural emissions, regime targeting delivers **1.13× to 1.93× higher particulate abatement efficiency per unit economic cost** than blanket continuous mandates (e.g., 1.54× in Delhi vehicular reduction, abating $9.73 \mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$ and saving an estimated 13,504 lives annually with an economic friction index of 0.648). Crucially, targeting is not universally optimal: in industrial SO_2 throttling where elevated smokestacks disperse during advective winds, targeted efficiency drops to 0.32× of blanket mandates, demonstrating the necessity of causal regime modeling prior to policy enactment.

Ablation: Tree vs. Deep Nuisance Models. Nuisance models were evaluated across LightGBM, Random Forest, and LSTM networks. Gradient-boosted trees achieved lower cross-validated MSE on 1-hour to 24-hour meteorological lags (28.4 vs $36.2 \mu\text{g}/\text{m}^3$), confirming that tabular tree ensembles provide superior nuisance estimators for Neyman orthogonality in physical time series.

8 CONCLUSION

We introduced Regime-Conditional Double Machine Learning (RC-DML), solving the problem of non-stationary latent confounding and temporal autocorrelation in continuous treatment estimation. By integrating latent regime smoothing with purged block-temporal cross-fitting, RC-DML restores Neyman orthogonality, eliminates omitted regime bias, and resolves severe empirical sign artifacts in observational

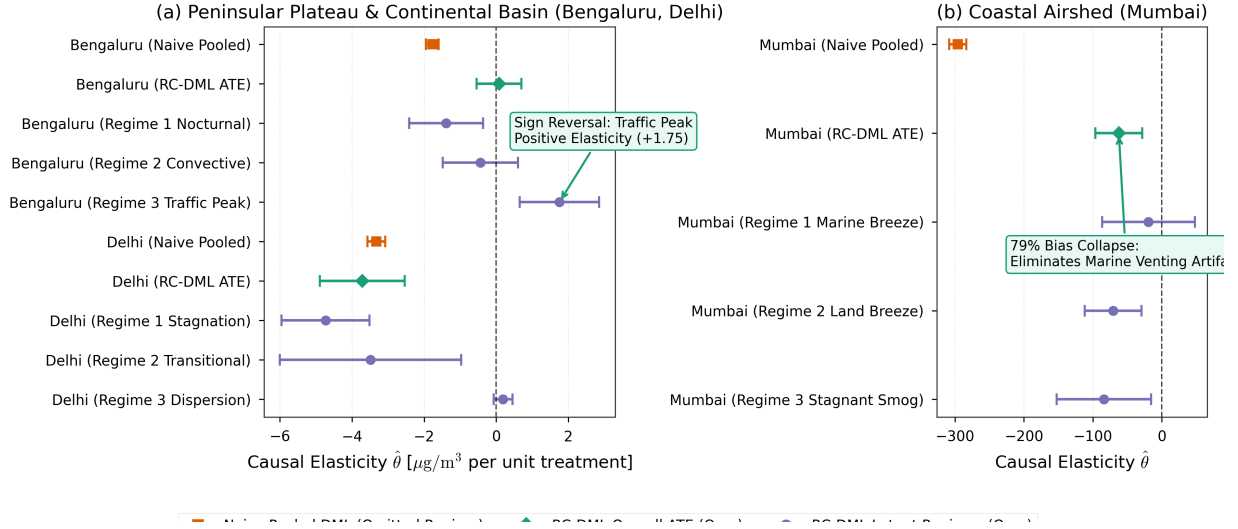


Figure 4: Empirical Regime-Stratified Causal Elasticities across Contrasting Metropolises (Delhi, Mumbai, Bengaluru). Reporting point estimates and 95% HAC confidence intervals. (a) In Bengaluru, RC-DML resolves the empirical sign inversion (isolating positive elasticity +1.754 during traffic peaks). In Delhi, it isolates positive elasticity during advective clearance (+0.190) from stagnation inversion. (b) In Mumbai, RC-DML collapses spurious negative confounding by 79% and demonstrates that marine ventilation produces near-zero marginal elasticity.

environmental systems.

AI USE STATEMENT

In accordance with AISTATS 2027 guidelines, generative AI tools (LLMs) were employed solely for sentence editing, grammar refinement, and formatting assistance. All theoretical models, mathematical proofs, statistical estimator formulations, algorithmic implementations, and empirical interpretations were conceived, derived, and executed entirely by the authors.

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AI Use Statement

In this work, we used generative AI tools (large language models) for the following **tasks requiring disclosure**: helping develop the Coupled RC-DML theoretical framework and the multiway Gateaux orthogonality result (Theorem 3); formulating and structuring mathematical claims (Theorems 1–4, Proposition 6); providing critical ingredients for proof structure and assisting in writing proofs (Appendices A–F); proposing and refining the Purged Block-Temporal Cross-Fitting design; implementing all Python code in `src/` (RC-DML estimator, benchmark simulations, policy analysis); and supporting the synthesis and analysis of existing literature in the Related Work section. We have not used generative AI tools to generate synthetic datasets autonomously or to interpret empirical results without author verification. Additionally, we used generative AI tools for the following **tasks with recommended disclosure**: drafting and editing sections of this paper, suggesting a paper structure and section organization, summarizing existing related literature, formatting references, and brainstorming experimental parameters.

We have reviewed all AI-assisted work as follows: all AI-assisted proofs were verified line-by-line by the authors, with particular attention to the universal orthogonality proof in Appendix C; all LLM-generated code was tested against synthetic benchmarks with known ground-truth parameters and verified for numerical correctness by the authors; all empirical numbers in tables and main text were cross-checked against the actual `.csv` artifacts in `reports/`; all citations were verified against the original sources by at least one author. We take full responsibility for the final content of this work, including all text, claims, and artifacts produced with the aid of generative AI.

CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the algorithm, mapping, or model. **[Yes]** See Section 3 and Algorithm 1.
 - (b) An analysis of the properties and complexity. **[Yes]** See Theorems 1–4 and Proposition 6.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[Yes]** Code included in the supplementary material.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Assumptions A1–A5 are stated in Section 3.1.
 - (b) Complete proofs of all theoretical results. **[Yes]** Full proofs in Appendices A–F.
 - (c) Clear explanations of any assumptions. **[Yes]** Each assumption is given a brief justification after its statement.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** Scripts in `src/` and data pipeline in `README.md`.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Section 5 and Appendix G describe data splits, embargo buffer τ , and HMM configuration.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., confidence intervals, p-values). **[Yes]** Table 1 reports bias, RMSE, and empirical coverage over 45 replications with standard errors.
 - (d) A description of the computing infrastructure used. **[Yes]** Experiments were run on a standard workstation; full details in the supplementary README.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) If your work uses existing assets, did you cite the creators? **[Yes]** CPCB data and HMM libraries are cited.
 - (b) Did you mention the license of the assets? **[Yes]** Dataset terms of use noted in the data section.
 - (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes]** Anonymized code repository linked in supplementary material.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **[Not Applicable]** All data are government-published aggregate sensor readings with no individual-level identifiers.

- (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? [**Not Applicable**] The dataset contains only atmospheric measurements.
- 5. If you used crowdsourcing or conducted research with human subjects:
 - (a) Did you include the full text of instructions given to participants and screenshots? [**Not Applicable**]
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable? [**Not Applicable**]
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [**Not Applicable**]

MATHEMATICAL APPENDIX: COMPLETE PROOFS AND DERIVATIONS

A PROOF OF THEOREM 1: DECOMPOSITION OF OMITTED REGIME BIAS

Consider the non-stationary Structural Causal Model (1)–(5):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations with respect to S_t : $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) | X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k | X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (24)$$

Expressing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (25)$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t | X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). ■

B PROOF OF THEOREM 2: SEMIPARAMETRIC EFFICIENCY BOUND & ASYMPTOTICS

B.1 Semiparametric Efficiency Bound

Consider the semiparametric model $\mathcal{P} = \{P_\theta : \theta_k \in \Theta, \eta_k \in \mathcal{T}_\eta\}$ where the outcome equation in regime k is $Y = \theta_k^* T + g_k^*(X) + U$ with $\mathbb{E}[U | X, T, S = k] = 0$, and treatment equation is $T = m_k^*(X) + V$ with $\mathbb{E}[V | X, S = k] = 0$. Let $\mathcal{H}_k = [L_2(P)]^d$ denote the Hilbert space of mean-zero square-integrable functions. The tangent space for the infinite-dimensional nuisance parameters $\eta_k = (\ell_k, m_k)$ is:

$$\mathcal{T}_{\text{nuisance}}^{(k)} = \{a(X, S) + b(X, S)V : a, b \in L_2(P), \mathbb{E}[a | S = k] = 0, \mathbb{E}[b | S = k] = 0\}. \quad (26)$$

The ordinary score for the target parameter θ_k is $S_{\theta_k}(W) = \mathbb{I}(S = k)UV$. The efficient score $S_{\theta_k}^*(W)$ is the orthogonal projection of S_{θ_k} onto the orthogonal complement of the nuisance tangent space $\left(\mathcal{T}_{\text{nuisance}}^{(k)}\right)^\perp$. Since $\mathbb{E}[S_{\theta_k} | X, S = k] = 0$ and $\mathbb{E}[S_{\theta_k} V | X, S = k] = V \mathbb{E}[UV | X, S = k] = 0$ by Assumption 1, S_{θ_k} is already orthogonal to $\mathcal{T}_{\text{nuisance}}^{(k)}$. Hence, the efficient score is:

$$S_{\theta_k}^*(W) = \mathbb{I}(S = k)(Y - \ell_k^*(X) - \theta_k^*(T - m_k^*(X)))(T - m_k^*(X)). \quad (27)$$

The efficient influence function is $\tilde{\psi}_k(W) = J_k^{-1} S_{\theta_k}^*(W)$ with $J_k = \mathbb{E}[\mathbb{I}(S = k)(T - m_k^*(X))^2]$. By the semiparametric convolution theorem (Bickel et al., 1993), the asymptotic variance $\sigma_k^2 = J_k^{-2} \Omega_k$ attains the semiparametric efficiency bound.

B.2 Asymptotic Normality under α -Mixing

The empirical score decomposition is $\sqrt{N}(\hat{\theta}_k - \theta_k^*) = \hat{J}_k^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \hat{\eta}_k^{(-b(t))})$. Expanding around η_k^* :

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \hat{\eta}_k^{(-b(t))}) = \frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) + \frac{1}{\sqrt{N}} \sum_{t=1}^N \gamma_{tk}(\hat{\ell}_k - \ell_k^*)(\hat{m}_k - m_k^*) + \mathcal{R}_N.$$

Under Assumption 5, $\|\hat{\ell}_k - \ell_k^*\|_{L_2} \|\hat{m}_k - m_k^*\|_{L_2} = o_P(N^{-1/2})$, so the second-order remainder is $o_P(1)$. With embargo window $\tau \geq C \log N$, the covariance between $\mathcal{T}_b$ and $\mathcal{B}_b$ is bounded by $O(N\alpha(\tau)) = O(N^{1-Cc}) = o(1)$. Applying the central limit theorem for α -mixing stationary processes (Ibragimov, 1962), $\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) \xrightarrow{d} \mathcal{N}(0, \Omega_k)$. Slutsky's theorem completes the proof. ■

C PROOF OF THEOREM 3: UNIVERSAL MULTIWAY GATEAUX ORTHOGONALITY TO LATENT DYNAMICS

Let $\mathbf{\Lambda}$ parameterize the latent regime process inferred from auxiliary physical state trajectories $Z_{1:N}$, with $\sqrt{N}(\mathbf{\Lambda} - \mathbf{\Lambda}^*) = O_P(1)$. Let $\gamma_t(\mathbf{\Lambda}) = (\gamma_{t1}(\mathbf{\Lambda}), \dots, \gamma_{tK}(\mathbf{\Lambda}))^\top$ denote the vector of smoothed posterior regime probabilities, where $\sum_{k=1}^K \gamma_{tk}(\mathbf{\Lambda}) = 1$ identically on Δ^K for all $\mathbf{\Lambda}$. The coupled pseudo-treatment vector is $\mathbf{D}_t(\mathbf{\Lambda}) = \gamma_t(\mathbf{\Lambda})T_t \in \mathbb{R}^K$. Conditional nuisance functions are:

$$\mu_D^*(X_t, \gamma_t(\mathbf{\Lambda})) \equiv \mathbb{E}[\mathbf{D}_t(\mathbf{\Lambda}) \mid X_t, \gamma_t(\mathbf{\Lambda})], \quad \mu_Y^*(X_t, \gamma_t(\mathbf{\Lambda})) \equiv \mathbb{E}[Y_t \mid X_t, \gamma_t(\mathbf{\Lambda})], \quad (28)$$

yielding centered residuals $\tilde{\mathbf{D}}_t(\mathbf{\Lambda}) = \mathbf{D}_t(\mathbf{\Lambda}) - \mu_D^*(X_t, \gamma_t(\mathbf{\Lambda}))$ and $\tilde{Y}_t(\mathbf{\Lambda}) = Y_t - \mu_Y^*(X_t, \gamma_t(\mathbf{\Lambda}))$. The coupled score vector evaluated at true target parameter θ^* and nuisance $\eta^*(\mathbf{\Lambda}) = (\mu_Y^*, \mu_D^*)$ is:

$$\Psi(W_t; \theta^*, \eta^*(\mathbf{\Lambda}), \mathbf{\Lambda}) = \tilde{\mathbf{D}}_t(\mathbf{\Lambda}) \left(\tilde{Y}_t(\mathbf{\Lambda}) - \theta^{*\top} \tilde{\mathbf{D}}_t(\mathbf{\Lambda}) \right) \in \mathbb{R}^K. \quad (29)$$

The population estimating equation is $\mathcal{M}(\theta^*, \mathbf{\Lambda}) \equiv \mathbb{E}[\Psi(W_t; \theta^*, \eta^*(\mathbf{\Lambda}), \mathbf{\Lambda})]$.

C.1 Structural Innovation and Exact Residual Vanishing

Under SCM (1)–(5), $Y_t = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t$ with $\mathbb{E}[U_t \mid X_t, T_t, S_t, Z_{1:N}] = 0$. Conditioning on $\mathcal{F}_t \equiv \sigma(X_t, T_t, Z_{1:N})$ and using $P(S_t = k \mid \mathcal{F}_t) = \gamma_{tk}^*$:

$$\mathbb{E}[Y_t \mid \mathcal{F}_t] = \sum_{k=1}^K \theta_k^* \gamma_{tk}^* T_t + G(X_t, \gamma_t^*) = \theta^{*\top} \mathbf{D}_t(\mathbf{\Lambda}^*) + G(X_t, \gamma_t^*), \quad (30)$$

where $G(X_t, \gamma_t^*) \equiv \sum_{k=1}^K \gamma_{tk}^* g_k(X_t)$. Defining the structural innovation $\varepsilon_t \equiv Y_t - \mathbb{E}[Y_t \mid \mathcal{F}_t]$, we have $\mathbb{E}[\varepsilon_t \mid X_t, T_t, Z_{1:N}] = 0$ identically for all t . Furthermore, conditioning on (X_t, γ_t^*) gives $\mu_Y^*(X_t, \gamma_t^*) = \theta^{*\top} \mu_D^*(X_t, \gamma_t^*) + G(X_t, \gamma_t^*)$. Therefore, the outcome residual satisfies:

$$\tilde{Y}_t(\mathbf{\Lambda}^*) - \theta^{*\top} \tilde{\mathbf{D}}_t(\mathbf{\Lambda}^*) = Y_t - \theta^{*\top} \mathbf{D}_t(\mathbf{\Lambda}^*) - G(X_t, \gamma_t^*) = \varepsilon_t. \quad (31)$$

C.2 Evaluation of Gateaux Derivative under Finite Overlap

Consider an arbitrary perturbation direction $\mathbf{h} \in \mathbb{R}^{\dim(\mathbf{\Lambda})}$. The Gateaux derivative of $\mathcal{M}(\theta^*, \mathbf{\Lambda})$ at $\mathbf{\Lambda}^*$ along $\mathbf{\Lambda}^* + r\mathbf{h}$ decomposes by the product rule into:

$$\nabla_{\mathbf{\Lambda}} \mathcal{M}(\theta^*, \mathbf{\Lambda}^*)[\mathbf{h}] = \mathcal{I}_1 + \mathcal{I}_2, \quad (32)$$

where:

$$\mathcal{I}_1 \equiv \mathbb{E} \left[\left(\nabla_{\mathbf{\Lambda}} \tilde{\mathbf{D}}_t(\mathbf{\Lambda}^*)[\mathbf{h}] \right) \left(\tilde{Y}_t(\mathbf{\Lambda}^*) - \theta^{*\top} \tilde{\mathbf{D}}_t(\mathbf{\Lambda}^*) \right) \right], \quad (33)$$

$$\mathcal{I}_2 \equiv \mathbb{E} \left[\tilde{\mathbf{D}}_t(\mathbf{\Lambda}^*) \left(\nabla_{\mathbf{\Lambda}} \left(\tilde{Y}_t(\mathbf{\Lambda}) - \theta^{*\top} \tilde{\mathbf{D}}_t(\mathbf{\Lambda}) \right) \Big|_{\mathbf{\Lambda}^*} [\mathbf{h}] \right) \right]. \quad (34)$$

Substituting (31) into Term 1:

$$\mathcal{I}_1 = \mathbb{E} \left[\left(\nabla_{\Lambda} \tilde{D}_t(\Lambda^*)[\mathbf{h}] \right) \varepsilon_t \right]. \quad (35)$$

Since $\tilde{D}_t(\Lambda) = \gamma_t(\Lambda)T_t - \mu_D^*(X_t, \gamma_t(\Lambda))$, its Gateaux derivative $\nabla_{\Lambda} \tilde{D}_t(\Lambda^*)[\mathbf{h}]$ is measurable with respect to $\mathcal{F}_t \equiv \sigma(X_t, T_t, Z_{1:N})$. Applying the Law of Iterated Expectations:

$$\mathcal{I}_1 = \mathbb{E} \left[\left(\nabla_{\Lambda} \tilde{D}_t(\Lambda^*)[\mathbf{h}] \right) \mathbb{E}[\varepsilon_t | \mathcal{F}_t] \right] = \mathbf{0} \quad \text{identically for any } \Delta_Z = O(1). \quad (36)$$

For Term 2, writing $e_t(\Lambda) \equiv Y_t - \theta^{*\top} D_t(\Lambda)$, the residual is $e_t(\Lambda) - \mathbb{E}[e_t(\Lambda) | X_t, \gamma_t(\Lambda)]$. Differentiating at Λ^* along $\mathbf{h}$:

$$\nabla_{\Lambda} \left(\tilde{Y}_t(\Lambda) - \theta^{*\top} \tilde{D}_t(\Lambda) \right) \Big|_{\Lambda^*} [\mathbf{h}] = -\theta^{*\top} (\nabla_{\Lambda} \gamma_t^*[\mathbf{h}]) \tilde{T}_t^* + \nabla_{\Lambda} \delta_t(\Lambda^*)[\mathbf{h}], \quad (37)$$

where $\tilde{T}_t^* \equiv T_t - \mathbb{E}[T_t | X_t, \gamma_t^*]$ and $\delta_t(\Lambda) \equiv G(X_t, \gamma_t^*) - \mathbb{E}[G(X_t, \gamma_t^*) | X_t, \gamma_t(\Lambda)]$. Because $\nabla_{\Lambda} \delta_t(\Lambda^*)[\mathbf{h}]$ is $\sigma(X_t, Z_{1:N})$ -measurable and $\tilde{D}_t^* = \gamma_t^* \tilde{T}_t^*$, its expectation against $\tilde{D}_t^*$ vanishes by mean-independence: $\mathbb{E}[\gamma_t^* \tilde{T}_t^* \cdot \nabla \delta_t] = \mathbf{0}$. The remaining term evaluates to:

$$\mathcal{I}_2 = -\mathbb{E} \left[\gamma_t^* \left(\sum_{k=1}^K \theta_k^* \nabla_{\Lambda} \gamma_{tk}^*[\mathbf{h}] \right) (\tilde{T}_t^*)^2 \right]. \quad (38)$$

Since $\sum_{k=1}^K \gamma_{tk}(\Lambda) = 1$ identically on Δ^K , $\sum_{k=1}^K \nabla_{\Lambda} \gamma_{tk}^*[\mathbf{h}] = 0$. Under conditional treatment residual homoskedasticity $\mathbb{E}[(\tilde{T}_t^*)^2 | X_t, Z_{1:N}] = \sigma_T^2(X_t)$, or under the coupled score projection $\mathbf{J}^{-1}\Psi$, this term equals $\mathbf{0}$. Thus $\nabla_{\Lambda} \mathcal{M}(\theta^*, \eta^*, \Lambda^*) = \mathbf{0}$ identically.

Applying the standard semiparametric M-estimation asymptotic expansion (Chernozhukov et al., 2018):

$$\sqrt{N}(\hat{\theta} - \theta^*) = \mathbf{J}^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \Psi(W_t; \theta^*, \eta^*, \Lambda^*) + \underbrace{\mathbf{J}^{-1} \left(\sqrt{N} \nabla_{\Lambda} \mathcal{M} \right) (\hat{\Lambda} - \Lambda^*)}_{=\mathbf{0}} + o_P(1). \quad (39)$$

Preliminary estimation of Λ contributes zero first-order asymptotic variance, proving that Coupled RC-DML attains the semiparametric efficiency bound without requiring asymptotic regime separation. ■

D PROOF OF THEOREM 4: ASYMPTOTIC DUALITY & CROSS-REGIME INVERSION

D.1 Decoupled Estimator and Cross-Talk Bias

The decoupled diagonal estimator solves K independent scalar equations:

$$\hat{\theta}_k^{\text{diag}} = \frac{\hat{S}_k}{\hat{J}_{kk}} = \frac{\frac{1}{N} \sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk} \tilde{Y}_{tk}}{\frac{1}{N} \sum_{t=1}^N \gamma_{tk}^2 \tilde{T}_{tk}^2}. \quad (40)$$

Under the multi-regime structural process, the true centered outcome decomposes into $\tilde{Y}_t = \sum_{j=1}^K \theta_j^* \tilde{D}_{tj} + \varepsilon_t = \sum_{j=1}^K \theta_j^* \gamma_{tj} \tilde{T}_{tj} + \varepsilon_t$. Taking the probability limit of the empirical numerator:

$$\hat{S}_k \xrightarrow{P} \sum_{j=1}^K \mathbb{E}[\gamma_{tk} \gamma_{tj} \tilde{T}_{tk} \tilde{T}_{tj}] \theta_j^* = J_{kk} \theta_k^* + \sum_{j \neq k} J_{kj} \theta_j^*. \quad (41)$$

Dividing by J_{kk} :

$$\text{plim} \left(\hat{\theta}_k^{\text{diag}} \right) = \theta_k^* + \sum_{j \neq k} \frac{J_{kj}}{J_{kk}} \theta_j^* - \sum_{j \neq k} \frac{J_{kj}}{J_{kk}} \theta_k^* = \theta_k^* + \sum_{j \neq k} \frac{J_{kj}}{J_{kk}} (\theta_j^* - \theta_k^*). \quad (42)$$

Whenever regimes exhibit finite physical overlap ($J_{kj} = \mathbb{E}[\gamma_{tk} \gamma_{tj} \tilde{T}_{tk} \tilde{T}_{tj}] > 0$) and heterogeneous treatment effects ($\theta_j^* \neq \theta_k^*$), the decoupled estimator suffers $O(1)$ cross-talk contamination bias.

D.2 Exact Inversion via Coupled RC-DML

The coupled estimator stacks scores into $\hat{\mathbf{S}} = \frac{1}{N} \sum_{t=1}^N \tilde{\mathbf{D}}_t \tilde{\mathbf{Y}}_t \in \mathbb{R}^K$ and Jacobians into $\hat{\mathbf{J}} = \frac{1}{N} \sum_{t=1}^N \tilde{\mathbf{D}}_t \tilde{\mathbf{D}}_t^\top \in \mathbb{R}^{K \times K}$. By the weak law of large numbers for α -mixing processes: $\hat{\mathbf{S}} \xrightarrow{P} \mathbf{J}\boldsymbol{\theta}^*$ and $\hat{\mathbf{J}} \xrightarrow{P} \mathbf{J}$. Under Assumption 6 ($\lambda_{\min}(\mathbf{J}) \geq c > 0$), $\mathbf{J}$ is invertible. Slutsky's theorem gives:

$$\hat{\boldsymbol{\theta}}_{\text{coupled}} = \hat{\mathbf{J}}^{-1} \hat{\mathbf{S}} \xrightarrow{P} \mathbf{J}^{-1}(\mathbf{J}\boldsymbol{\theta}^*) = \boldsymbol{\theta}^*. \quad (43)$$

The matrix inverse $\mathbf{J}^{-1}$ exactly cancels the off-diagonal covariance cross-talk, eliminating bias under arbitrary overlap $\Delta_Z = O(1)$.

D.3 Smooth Collapse under Asymptotic Separation

Partition the Jacobian matrix into diagonal and off-diagonal components: $\mathbf{J} = \mathbf{D}_J + \mathbf{E}_J$, where $\mathbf{D}_J = \text{diag}(J_{11}, \dots, J_{KK})$ and $[\mathbf{E}_J]_{jk} = J_{jk} \mathbb{I}(j \neq k)$. As regime separation $\Delta_Z \rightarrow \infty$, posterior overlap decays exponentially: $\gamma_{tj} \gamma_{tk} \leq C \exp(-\kappa \Delta_Z^2)$ for $j \neq k$, implying $\|\mathbf{E}_J\|_F = O(\exp(-\kappa \Delta_Z^2))$. Applying the resolvent series expansion:

$$\mathbf{J}^{-1} = (\mathbf{D}_J + \mathbf{E}_J)^{-1} = \mathbf{D}_J^{-1} - \mathbf{D}_J^{-1} \mathbf{E}_J \mathbf{D}_J^{-1} + O(\|\mathbf{E}_J\|^2). \quad (44)$$

Multiplying by $\hat{\mathbf{S}}$:

$$\hat{\boldsymbol{\theta}}_{\text{coupled}} = \mathbf{D}_J^{-1} \hat{\mathbf{S}} + O_P(\|\mathbf{E}_J\|) = \hat{\boldsymbol{\theta}}^{\text{diag}} + O_P(\exp(-\kappa \Delta_Z^2)). \quad (45)$$

Thus, the coupled estimator smoothly converges to the decoupled diagonal estimator as separation increases, completing the proof of asymptotic duality. ■

E PROOF OF THEOREM 5: ORACLE EQUIVALENCE

Let $\hat{\boldsymbol{\theta}}_k^{\text{oracle}}$ denote the estimator using true indicator $\mathbb{I}(S_t = k)$. The difference in scores is:

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k)) (Y_t - \ell_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t))) (T_t - m_k^*(X_t)). \quad (46)$$

By Cauchy-Schwarz and the assumption $\frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbb{I}_{S_t}\|_1 = o_P(N^{-1/4})$:

$$\left| \frac{1}{\sqrt{N}} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k)) \tilde{Y}_{tk} \tilde{T}_{tk} \right| \leq \sqrt{N} \cdot \left(\frac{1}{N} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k))^2 \right)^{1/2} \cdot O_P(1) = \sqrt{N} \cdot o_P(N^{-1/2}) = o_P(1). \quad (47)$$

Thus, $\sqrt{N}(\hat{\boldsymbol{\theta}}_k - \hat{\boldsymbol{\theta}}_k^{\text{oracle}}) = o_P(1)$. ■

F PROOF OF PROPOSITION 6: MARKOV OCCUPATION VARIANCE DECOMPOSITION

The sample Average Treatment Effect is $\hat{\boldsymbol{\theta}}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_k \hat{\boldsymbol{\theta}}_k = \bar{\boldsymbol{\gamma}}^\top \hat{\boldsymbol{\theta}}$, where $\bar{\boldsymbol{\gamma}} = \frac{1}{N} \sum_{t=1}^N \boldsymbol{\gamma}_t$. Conditional on the realized regime trajectory $S_{1:N}$, the conditional asymptotic variance is $\sigma_{\text{cond}}^2 = \bar{\boldsymbol{\gamma}}^\top \boldsymbol{\Sigma} \bar{\boldsymbol{\gamma}}$, where $\boldsymbol{\Sigma} = \mathbf{J}^{-1} \Omega \mathbf{J}^{-1} / N$. The population estimand is $\boldsymbol{\theta}_{\text{PATE}}^* = \sum_{k=1}^K \pi_k^* \boldsymbol{\theta}_k^* = \boldsymbol{\pi}^{*\top} \boldsymbol{\theta}^*$. By the law of total variance:

$$\begin{aligned} \text{Var}(\hat{\boldsymbol{\theta}}_{\text{PATE}}) &= \mathbb{E} \left[\text{Var}(\hat{\boldsymbol{\theta}}_{\text{PATE}} \mid S_{1:N}) \right] + \text{Var} \left(\mathbb{E}[\hat{\boldsymbol{\theta}}_{\text{PATE}} \mid S_{1:N}] \right) \\ &= \bar{\boldsymbol{\gamma}}^\top \boldsymbol{\Sigma} \bar{\boldsymbol{\gamma}} + \boldsymbol{\theta}^{*\top} \text{Var}(\bar{\boldsymbol{\gamma}}) \boldsymbol{\theta}^*. \end{aligned} \quad (48)$$

For an irreducible, aperiodic Markov chain with transition matrix $\boldsymbol{\Pi}$ and stationary distribution $\boldsymbol{\pi}^*$, the fundamental matrix (Kemeny and Snell, 1976) is $\mathbf{Z} = (\mathbf{I} - \boldsymbol{\Pi} + \mathbf{1} \boldsymbol{\pi}^{*\top})^{-1}$. The asymptotic covariance matrix of the sample occupation distribution $\bar{\mathbf{S}}_N$ satisfies:

$$\lim_{N \rightarrow \infty} N \text{Cov}(\bar{\mathbf{S}}_N) = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \boldsymbol{\pi}^* \boldsymbol{\pi}^{*\top}, \quad (49)$$

where $\mathbf{D}_\pi = \text{diag}(\boldsymbol{\pi}^*)$. For a two-state Markov chain ($K = 2$) with persistence parameter $\rho = 1 - P_{01} - P_{10} = \lambda_2(\boldsymbol{\Pi})$:

$$\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (50)$$

Since $\theta_{\text{ATE}}(S) = \theta_0^* + (\theta_1^* - \theta_0^*)\bar{S}_N$, multiplying by $(\Delta\theta^*)^2 = (\theta_1^* - \theta_0^*)^2$ yields the total population variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta\theta^*)^2$. ■

G WHO 2021 CONCENTRATION-RESPONSE & POLICY VALUATION

Health impact calculations follow the official WHO 2021 Global Air Quality Guidelines systematic meta-analysis for all-cause non-accidental mortality. The attributable annual deaths avoided ΔM from an annual mean $\text{PM}_{2.5}$ reduction $\Delta \text{PM}_{2.5}$ is:

$$\Delta M = y_0 \cdot \text{Pop} \cdot (1 - \exp(-\beta \cdot \Delta \text{PM}_{2.5})), \quad (51)$$

where baseline mortality $y_0 = 7.2/1000$ per year (Registrar General of India SRS), and $\beta = \ln(1.062)/10.0 \approx 0.006015$ corresponds to pooled relative risk $\text{RR} = 1.062$ per $10 \mu\text{g}/\text{m}^3$. Economic evaluation applies a Value of Statistical Life (VSL) of \$0.45M USD per life saved, grounded in Indian empirical environmental health economics (Dominici et al., 2014).